Adaptive Fuzzy Tracking Control for Uncertain Nonlinear Systems With Unknown Control Gain Functions via Intermittent Output

Xinjun Wang, Shenghang Liu, Xin Wang, Ben Niu, Wenqi Zhou, and Xinmin Song

***Abstract*—Based on output triggering, an adaptive prescribed-time tracking control strategy is proposed for a class of uncertain strict-feedback nonlinear systems with unknown control gains in this article. The nondifferentiability of the virtual control signals is identified as the most prominent design difficulty in this research. In order to solve the above difficulty, a new fuzzy state observer is built by using triggered output signal and fuzzy logic systems (FLSs), which in turn generates alternative continuous states. Simultaneously, the estimated signals are utilized to design virtual control signals, making certain that the virtual control signals have a well-defined first derivative. On this basis, by introducing a time-varying constraint function, a new adaptive prescribed-time fuzzy controller is constructed, so that the controller can still be applied to continuously operating systems after the predefined time. Additionally, we introduce the command filtering technique to mitigate repeated differentiation of virtual control signals during the backstepping design process. Combining the constructed logarithmic Lyapunov functions and bounded control technique with backstepping design, it is possible to guarantee that the established adaptive prescirbed-time event-triggered control method satisfies the following: 1) within the predefined time, the tracking error converges to the user-specified region and 2) the full range of signals involved in the closed-loop system is kept bounded.**

At last, the results of the single-link arm simulation example verify the reasonableness and effectiveness of the established control scheme.

***Index Terms*—Adaptive tracking control, fuzzy state observer, fuzzy-logic systems (FLS), output triggering.**

I. Introduction

With the development of the times, the system functions are increasing, more and more data need to be transmitted, and the limited communication resources cannot satisfy the status quo, which prompts researchers to search for more effective and reasonable methods to reduce the communication cost among the components of the control system. Over the past few decades, the event-triggered control strategies have become increasingly popular [1], [2], [3]. An adaptive event-triggered controller was constructed for a category of output feedback uncertain nonlinear systems based on adaptive dynamic gain [4]. Distinct from the conventional adaptive event-triggered control methodologies, the control scheme proposed by [5] not only ensured simultaneous triggering of both the parameter estimator and the controller, but also achieved asymptotic convergence of the stabilization error. It is important to note that the above literatures mainly focused on event-triggered strategies from the controller-to-actuator channel. In the industrial network environments, the application of event-triggered mechanisms to the sensor-to-controller channel not only saves communication resources in that channel but also has significant economic value. Regarding the first-order uncertain nonlinear systems, the distributed consensus tracking control problem based on event-triggering mechanism was investigated in [6]. In [7] and [8], the issues of event-triggered control in the sensor-to-controller channel were investigated in both strict-feedback as well as nonstrict-feedback systems, respectively. Nevertheless, the application of the output triggering mechanism causes the output signal received by the controller to become discontinuous, leading to nondifferentiable virtual control signals. This poses a significant challenge to backstepping control design.

In addition, in certain fields, such as space docking, missile interception, and drone formations, achieving ideal tracking accuracy within a short timeframe is essential [9]. On the basis of the backstepping technique, an adaptive fuzzy control scheme was constructed to tackle the finite-time tracking problem for nonlinear pure-feedback systems [10]. For stochastic nonlinear systems, a finite-time command filter-based adaptive fuzzy tracking control scheme was proposed by means of event-triggered mechanism to ensure that the output y tracks the given desired trajectory signal in the finite time [11]. Nevertheless, the convergence time and tracking accuracy of the finite time control method designed in the

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above study are dependent on initial conditions and other design parameters. In order to overcome this difficulty, the prescribed-time control strategies for nonlinear systems were proposed by some researchers [12]. Compared with other control methods, the prescribed-time control strategy can meet the user's predesigned convergence time and tracking accuracy requirements, which greatly improves the practicality of the control schemes. Based on Lyapunov–Krasovskii functional scheme and the prescribed-time adjustment function, a novel definition of prescribed-time Lyapunov stability was built [13]. In [14], the prescribed-time control of time-varying nonlinear systems was achieved through using a time-transition method. Based on the definition of the general prescribed-time function, it is not difficult to see that as the time t approaches the predefined time T, it may lead to the issue of infinite gain. In order to conquer this obstacle, [15] proposed a control strategy using partial tracking error constraints based on the performance function for nonlinear systems. The above literatures generally assume that the initial set satisfies $-\eta(0) \leq z_1(0) \leq \eta(0)$, in which $\eta(t)$ is the prescribed-time constraint function as well as $z_1(0)$ is the tracking error of the system. However, this approach cannot be applied when the system's preliminary values are infinite, which urgently demands that we consider more general assumptions.

It is well known that uncertain nonlinear terms widely exist in industrial nonlinear system models, how to deal with these nonlinear terms safely and effectively has become a popular research area. Recently, the ability of radial basis function neural networks (RBF NNs) and fuzzy logic systems (FLSs) to approximate unknown smooth nonlinear functions has been recognized by more and more scholars [16], [17]. It is combined with the backstepping technique, which in turn deals with the intelligent control of nonlinear systems. To overcome the difficulties associated with nonstrict-feedback structure, a new adaptive neural network backstepping output feedback control method was proposed for nonlinear nonstrict-feedback systems [18]. In [19], a new adaptive event-triggered asymptotically consistent tracking control scheme for nonlinear multiagent systems (MASs) was proposed by combining RBF NNs with improved dynamical surface control technique. For the nonlinear cyber-physical systems with backlashlike hysteresis, an improved event-triggered adaptive fuzzy control scheme was built in [20]. For nonstrict-feedback nonlinear systems, a finite-time adaptive prescribed performance fuzzy control problem was investigated in [21].

Inspired by the aforementioned findings, this study aims to tackle the challenge of adaptive prescribed-time event-triggered tracking control for uncertain nonlinear systems with unknown control gains. The key contributions of this work are summarized as follows.

1) It is important to note that the event-triggered control algorithms studied in [4], [5], [22], [23], [24], [25], [26], and [27] only considered the triggered control signal. This is due to the fact that the control signal only appears in the final design step and does not require differentiation, it does not pose any difficulties in the backstepping design process. In this article, the event-triggered control strategy proposed is characterised by: a) the more difficult output triggering is considered. By considering output triggering directly, the impacts of error introduced by replacing continuous signals with triggered signals are avoided, simplifying the control design process; b) overcome the difficulty of nondifferentiability of virtual control signals caused by output triggering and c) avoid the transmission of unnecessary sample information, reducing communication burden.
2) In order to enhance the tracking performance of the considered nonlinear systems, unlike the schemes proposed in [28], [29], [30], and [31], this article establishes a novel adaptive prescribed-time fuzzy controller, which allows the tracking error to converge to the user-specified preset region within the predefined time. Moreover, the predefined time and tracking accuracy are decoupled from initial conditions and other design parameters, allowing the user to design their values according to the actual engineering context. Furthermore, the conventional assumption regarding the initial values of the system in [15] and [32] is extended to a more general assumption that the initial values can be infinite.
3) Through the use of the triggered output signal and FLSs, a new state observer is established, resulting in substitute continuous states. The logarithmic Lyapunov functions are created during the backstepping design process, and a new adaptive control strategy is constructed by introducing the command filtering technique and bounded control design technique.

II. Problem Formulation and Technical Tools

A. Description of the System

The model described below can be used to represent the strict-feedback nonlinear system that this work examines

$$\begin{cases} \dot{x}_k = \mathsf{g}_k(\bar{x}_k)x_{k+1} + f_k(\bar{x}_k), \ k = 1, \ldots, n-1 \\ \dot{x}_n = \mathsf{g}_n(\bar{x}_n)u + f_n(\bar{x}_n) \\ y = x_1, \end{cases} \tag{1}$$

where $y \in \Re$, $\bar{x}_k = [x_1, \ldots, x_k]^T \in \Re^k$ and $u \in \Re$ are the output, the state vectors of the system and input of the system, respectively. Here, we define $f_k(\diamond)$ and $\mathsf{g}_k(\diamond)$ as the unknown and smooth functions with $k = 1, \ldots, n$.

Assumption 1: There exist positive constants $\mathsf{g}_{k,\max}$ and $\mathsf{g}_{k,\min}$ so that $\mathsf{g}_{k,\max} \geq |\mathsf{g}_k(\bar{x}_k)| \geq \mathsf{g}_{k,\min}$ holds.

Remark 1: By Assumption 1, we can obtain the fact that the control gain functions $\mathsf{g}_k(\bar{x}_k)$ are strictly either positive or negative. The assumption can be found in much of the literatures [33], [34].

The main goal of this article is to only use the output signal after triggering to determine an appropriate state observer and develop an adaptive prescribed-time fuzzy controller, which aims to the following.

1) Achieve tracking control within the predefined time T, that is $\hat{y} \dashrightarrow y_r$.
2) Ensure that all signals involved in the closed-loop system are bounded.

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B. Event-Triggering Mechanism

In engineering applications, the actual communication capacity and bandwidth are limited. To save communication resources, the following event-triggering mechanism is constituted

$$\begin{aligned} \bar{y}(t) &= y\big(t_{y,k}\big), \quad t \in \big[t_{y,k}, t_{y,k+1}\big) \\ t_{y,k+1} &= \inf\big\{t > t_{y,k}{:}|y(t) - \bar{y}(t)| \geq \Lambda\big\} \end{aligned} \tag{2}$$

where $t_{y,k}$ is defined as the triggering time instant for the sensor output with $k = 0, 1, \ldots$. Λ is the designed positive constant and satisfies $\Lambda < \bar{\Lambda}$. Without losing any sense of generality, the first event is assumed to occur at the initial time. When $t \in [t_{y,k}, t_{y,k+1})$, the triggered output information $\bar{y}(t)$ is always kept as a constant $y(t_{y,k})$. When model (2) occurs, $\bar{y}(t) = y(t_{y,k+1})$ will be obtained.

Remark 2: The parameter Λ represents the threshold designed by the user, but the specific value for Λ needs to be selected based on the practical application specifications. In the simulation, a trial-and-error approach is applied. Determining an appropriate value for Λ is an important topic deserving further research.

C. The Fuzzy-Logic Systems

The FLSs are used to approximate the unknown continuous nonlinear terms in the system (1). For $\forall f(x)$ being a continuous function defined on the compact set $\Omega_x \in R^l$, there is

$$f(x) = \omega^{*T} S(x) + \epsilon(x) \tag{3}$$

where ω^* is the unknown ideal approximating weight matrix and let $S(x) = [S_1(x), \ldots, S_l(x)]^T / \sum_{j=1}^{l} S_j(x)$ be the fuzzy basis function vector. In addition, $l > 1$ is the fuzzy rule and $\epsilon(x)$ is defined as the approximation error with $|\epsilon(x)| \leq \epsilon^*$. ϵ^* is the given accuracy, satisfying $\epsilon^* > 0$.

Lemma 1 [35]: For $w = [w_1(x), \ldots, w_l(x)]^T$ that satisfies the following inequalities:

$$\begin{aligned} \tanh^2(\|w\|) &\leq \|\mathrm{Tanh}(w)\|^2 \\ \|\mathrm{Tanh}(w)\| &\leq \|w\| \\ \frac{\|w\|}{\tanh(\|w\|)} &\leq \|w\| + 1 \end{aligned} \tag{4}$$

where $\mathrm{Tanh}(w) = [\tanh(w_1), \ldots, \tanh(w_l)]^T$.

III. Main Results

For more convenient design of the state observer directly, we convert the system (1) into the nonstrict-feedback system, which is modeled as follows:

$$\begin{cases} \dot{x}_k = x_{k+1} + \hbar_k(\bar{x}_{k+1}) \\ \dot{x}_n = \hbar_n(\bar{x}_n) + \mathsf{g}_n(\bar{x}_n)u \\ y = x_1 \end{cases} \tag{5}$$

where $k = 1, \ldots, n-1$ and

$$\begin{aligned} \hbar_k(\bar{x}_{k+1}) &= \mathsf{g}_k(\bar{x}_k)x_{k+1} - x_{k+1} + f_k(\bar{x}_k) \\ \hbar_n(\bar{x}_n) &= f_n(\bar{x}_n). \end{aligned} \tag{6}$$

A. Design of Novel State Observer

Through combining (5) and (6), it is clear that $\hbar_{k+1}(\bar{x}_k)$ and $\mathsf{g}_n(\bar{x}_n)$ are the unknown continuous nonlinear functions. Therefore, the FLSs are introduced such that $|\hbar_k(\bar{x}_{k+1}) - \omega_k^{*T} S(\bar{x}_{k+1})| \leq \epsilon_k$ and $|\mathsf{g}_n(\bar{x}_n) - \omega_g^{*T} S(\bar{x}_n)| \leq \epsilon_g$, where ω_k^* and ω_g^* are ideal weights, ω_k and ω_g are estimations of ω_k^* and ω_g^*. In the work of this article, we assume that $\omega_k^* \in \Omega_k = \{\omega_k^*{:}||\omega_k^*|| \leq M_k\}$, $\omega_g^* \in \Omega_g = \{\omega_g^*{:}||\omega_g^*|| \leq M_g\}$, $|\epsilon_k| \leq \bar{\epsilon}_k$ and $|\epsilon_g| \leq \bar{\epsilon}_g$ with $\bar{\epsilon}_k$ and $\bar{\epsilon}_g$ being unknown positive constants.

In order to remodel the unknown real states of system, using only the triggered output signal $\bar{y}$, a novel observer is proposed, which can be defined as

$$\begin{cases} \dot{\hat{\bar{x}}}_k = \hat{x}_{k+1} + \ell_k(\bar{y} - \hat{y}) + \hat{\hbar}_k\Big(\hat{\bar{x}}_{k+1}|\omega_k\Big) \\ \dot{\hat{x}}_n = \hat{\mathsf{g}}_n\Big(\hat{\bar{x}}_n|\omega_n\Big)u + \ell_n(\bar{y} - \hat{y}) \ + \hat{\hbar}_n\Big(\hat{\bar{x}}_n|\omega_n\Big) \\ \hat{y} = \hat{x}_1 \end{cases} \tag{7}$$

where $k = 1, \ldots, n-1$. $\hat{x}_k$ is the estimate of x_k, $\hat{\bar{x}}_k = [\hat{x}_1, \ldots, \hat{x}_n]^T$. $\mathcal{L} = \big[\ell_1, \ell_2, \ldots, \ell_n\big]^T$ with $\mathcal{L}$ being the design matrix and the matrix $\mathcal{A}$ is a strict Hurwitz matrix [the form of the matrix $\mathcal{A}$ will be given in (11)]. Further, there exist positive definite matrices $\mathcal{Q}$ and $\mathcal{P}$ such that

$$\mathcal{A}^T\mathcal{P} + \mathcal{P}\mathcal{A} = -2\mathcal{Q}. \tag{8}$$

Then, let $\varrho = x - \hat{x} = [\varrho_1, \varrho_2, \ldots, \varrho_n]^T$ be the estimate error, the following equations hold:

$$\begin{aligned} \dot{\varrho}_k = \ & \varrho_{k+1} - \hat{\hbar}_k\Big(\hat{\bar{x}}_{k+1}|\omega_k\Big) + \ell_k(y - \bar{y}) + \hbar_k(\bar{x}_{k+1}) \\ & -\ell_k\varrho_1 \\ \dot{\varrho}_n = \ & \hbar_n(\bar{x}_n) + \Big(\mathsf{g}_n(\bar{x}_n) - \hat{\mathsf{g}}_n\Big(\hat{\bar{x}}_n|\omega_n\Big)\Big)u - \hat{\hbar}_n\Big(\hat{\bar{x}}_n|\omega_n\Big) \\ & +\ell_n(y - \bar{y}) - \ell_n\varrho_1 \end{aligned} \tag{9}$$

where $k = 1, \ldots, n-1$. From (9), let $\varrho = [\varrho_1, \ldots, \varrho_n]$ and the estimation error ϱ can be rewritten as

$$\begin{aligned} \dot{\varrho} = \ & \mathcal{A}\varrho + \sum_{k=1}^{n-1} \mathcal{M}_k\Big(\hbar_k(\bar{x}_{k+1}) - \hat{\hbar}_k\Big(\hat{\bar{x}}_{k+1}|\omega_k\Big)\Big) \\ & + \mathcal{L}(y - \bar{y}) + \mathcal{M}_n\Big(\mathsf{g}_n(\bar{x}_n) - \hat{\mathsf{g}}_n\Big(\hat{\bar{x}}_n|\omega_n\Big)\Big)u \\ & + \mathcal{M}_n\hbar_n(\bar{x}_n) - \hat{\hbar}_n\Big(\hat{\bar{x}}_n|\omega_n\Big) \\ = \ & \mathcal{A}\varrho + \sum_{k=1}^{n} \mathcal{M}_k\sigma_k + \mathcal{M}_n\sigma_g u + \mathcal{L}(y - \bar{y}) \end{aligned} \tag{10}$$

where

$$\mathcal{A} = \begin{bmatrix} -\ell_1 & 1 & & \\ -\ell_2 & 0 & 1 & \\ \vdots & \vdots & & \ddots \\ -\ell_{n-1} & 0 & \ldots & 0 \ \ 1 \\ -\ell_n & 0 & \cdots & 0 \ \ 0 \end{bmatrix}, \mathcal{M}_k = \Big[\underbrace{0, \ldots, 1}_{k}, 0, 0, \ldots, 0\Big]^T \tag{11}$$

and $\sigma_k = -\hat{\hbar}_k(\hat{\bar{x}}_{k+1}|\omega_k) + \hbar_k(\bar{x}_{k+1})$, $\sigma_n = -\hat{\hbar}_n(\hat{\bar{x}}_n|\omega_n) + \hbar_n(\bar{x}_n)$, $\sigma_g = -\hat{\mathsf{g}}_n(\hat{\bar{x}}_n|\omega_n) + \mathsf{g}_n(\bar{x}_n)$ with $\sigma = [\sigma_1, \ldots, \sigma_n]^T$.

The following Lyapunov function V_0 is chosen

$$V_0 = \tfrac{1}{2}\varrho^T\mathcal{P}\varrho. \tag{12}$$

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Then, the derivative of V_0 is given as

$$\dot{V}_0 = \frac{1}{2}\varrho^T(\mathcal{P}\mathcal{A}+\mathcal{A}^T\mathcal{P})\varrho + \varrho^T\mathcal{P}\left(\sum_{k=1}^{n}\mathcal{M}_k\sigma_k + \mathcal{M}_n\sigma_g u\right) + \varrho^T\mathcal{P}\mathcal{L}(y-\bar{y}). \tag{13}$$

Using the FLSs, we can get the following:

$$\sigma_k = \omega_k^{*T}\left(S_k(\bar{x}_{k+1}) - S_k\left(\hat{\bar{x}}_{k+1}\right)\right) + \epsilon_k + \tilde{\omega}_k^T S_k\left(\hat{\bar{x}}_{k+1}\right)$$
$$\sigma_g = \omega_g^{*T}\left(S_g(\bar{x}_{k+1}) - S_g\left(\hat{\bar{x}}_{k+1}\right)\right) + \epsilon_g + \tilde{\omega}_g^T S_g\left(\hat{\bar{x}}_{k+1}\right).$$

Combining the Young's inequality and $0 < S(\diamond)_k^T S(\diamond)_k \le 1$, we have

$$\begin{aligned}
\varrho^T\mathcal{P}\mathcal{L}(y-\bar{y}) \le\ & \frac{c}{2}\|\varrho\|^2\|\mathcal{P}\|^2 + \frac{4}{c}\|\mathcal{L}\|^2\bar{\Lambda}^2 \\
\varrho^T\mathcal{P}\sum_{k=1}^{n}\mathcal{M}_k\sigma_k \le\ & \frac{5c}{8}\|\varrho\|^2\|\mathcal{P}\|^2 \\
& + \sum_{k=1}^{n}\left[\frac{2\epsilon_k^{*2}}{c} + \frac{4\left\|\omega_k^*\right\|^2}{c} + \frac{\|\tilde{\omega}_k\|^2}{c}\right] \\
\varrho^T\mathcal{P}\mathcal{M}_n\sigma_g u \le\ & \frac{c}{2}\|\varrho\|^2\|\mathcal{P}\|^2 + \frac{1}{c}\left\|\tilde{\omega}_g\right\|^2 u^2 \\
& + \left(\frac{4\left\|\omega_g^*\right\|^2}{c} + \frac{2\epsilon_g^{*2}}{c}\right)u^2
\end{aligned} \tag{14}$$

where c is a positive constant.

Bring (14) into (13), we can easily obtain

$$\dot{V}_0 \le -\rho_0\|\varrho\|^2 + \frac{1}{c}\sum_{i=1}^{n}\|\tilde{\omega}_i\|^2 + \left[\frac{1}{c}\left\|\tilde{\omega}_g\right\|^2 + \varsigma\right]u^2 + \Theta_0 \tag{15}$$

and

$$\begin{aligned}
\rho_0 &= \lambda_{\min}(\mathcal{Q}) - \frac{13c}{8}\|\mathcal{P}\|^2 \\
\Theta_0 &= \sum_{i=1}^{n}\left[\frac{4\left\|\omega_i^*\right\|^2}{c} + \frac{2\epsilon_i^{*2}}{c}\right] + \frac{4}{c}\|\mathcal{L}\|^2\bar{\Lambda}^2 \\
\varsigma &= \left(\frac{4\left\|\omega_g^*\right\|^2}{c} + \frac{2\epsilon_g^{*2}}{c}\right)
\end{aligned} \tag{16}$$

where ρ_0 is chosen to satisfy the condition $\rho_0 > 0$.

Remark 3: Due to the boundedness of $\tilde{\omega}_i$, $\tilde{\omega}_g$, and u are unknown, it cannot be concluded from (15) that the state observer (7) is stable. In order to address these issues, in the next section, the boundedness of $\tilde{\omega}_i$, $\tilde{\omega}_g$ can be ensured by employing the projection operator technique. Then, it can be seen from (15) that if ρ_0 is chosen to be larger enough and $\tilde{\omega}_i$, $\tilde{\omega}_g$ and u are bounded, we can obtain the observer estimation error is bounded.

B. Adaptive Prescribed-Time Control Design

By considering the predefined time $T \in (0,\infty)$, immediately after we introduce the preset tracking accuracy $m \in (0,\infty)$. The following time-varying constraint function model is obtained

$$\eta(t) = \begin{cases} m, & t > T \\ \left(\frac{1}{t}-\frac{1}{T}\right)^{2p} + m, & 0 \le t \le T \end{cases} \tag{17}$$

where p is a positive integer. In order to ensure continuous differentiability of $\eta^{(k)}(t)(k=1,\ldots,n)$, let p satisfy $2p > n+1$ with n representing the order of the system.

Define the tracking error as $z_1 = \hat{y} - y_r$. In order to attain the desired tracking target within the predefined time, adherence to the following tracking error constraint is ensured:

$$-\eta(t) \le z_1(t) \le \eta(t). \tag{18}$$

It is worth noting that, in conjunction with the characteristic of the scalar function η in (17), we can further derive from (18)

$$|z_1(t)| < m, \quad t \ge T. \tag{19}$$

Given the above considerations, it is straightforward to conclude that under the premise of satisfying constraint (18), the tracking error converges to user desired region.

From (17), it can be observed that there is no constraint imposed at the initial time, thus resulting in the constraint boundary being infinite at the initial moment. In view of this, we define the following constraint control strategy:

$$\psi(\vartheta(t)) = \begin{cases} -\left(\frac{\vartheta(t)}{d}-1\right)^{2p}+1, & 0 < \vartheta(t) \le d \\ 1, & \vartheta(t) > d \end{cases} \tag{20}$$

where

$$\vartheta(t) = b(\eta(t) - z_1(t))(\eta(t) + z_1(t)) = b[\eta^2(t) - z_1^2(t)]$$

b and d are positive constants that have been designed. Noting that $\psi(\vartheta(t))$ in (20) is the smooth composite function (including the case where $\vartheta(t) = d$). At the same time, the value of $\vartheta(t)$ can be mapped to the interval $(0,1]$ by using the function $\psi(\vartheta(t))$.

For better tracking performance, a coordinate transformation is constructed

$$\mathcal{Z}_1(t) = \frac{z_1(t)}{\psi(\vartheta(t))} = \frac{z_1(t)}{\psi(b(\eta^2(t) - z_1^2(t)))}. \tag{21}$$

The above $\mathcal{Z}_1$ has the following characteristics. 1) $\mathcal{Z}_1 = 0$ holds if and only if $z_1 = 0$ and 2) when $z_1 \to \eta(t)$ and $z_1 \to -\eta(t)$, $\mathcal{Z}_1 \to \infty$. To guarantee that $-\eta(t) < z_1 < \eta(t)$ always holds for all $t > 0$, we require $\mathcal{Z}_1 \in L_\infty$.

Remark 4: Depending on the characteristic of the constraint function $\eta(t)$, it is known that $\eta(0) = \infty$, thus we can get the initial time $-\eta(0) < z_1(0) < \eta(0)$ holds. In order to further satisfy the constraint performance, we are required to demonstrate that $-\eta(t) < z_1(t) < \eta(t)$ holds for $t \in (0,\infty)$. For convenience of proof, we first define $D(t) = [z_1(t)/\eta(t)]$. Assume that there exists a time instant $t = t_1$ so that $z_1(t) \le -\eta(t)$ or $z_1(t) \ge \eta(t)$ holds, thus $D(t) \ge 1$ or $D(t) \le -1$ is satisfied. Based on the theorem in [36], we can get that there has a time instant $0 < t_2 < t_1$ such that $D(t_2) = 1$ or $D(t_2) = -1$. Further, $\vartheta(t_2) = b(\eta^2(t_2) - z_1^2(t_2))$. Combining (20) and (21), we obtain $\psi(\vartheta(t_2)) = 0$ and $\mathcal{Z}_1(t_2) = \infty$, which in turn lead to a contradiction between the above result and

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the boundedness of $\mathcal{Z}_1(t)$. As a result of the above analysis, we can finally get $-1 < D(t) < 1$ and $-\eta(t) < z_1(t) < \eta(t)$ always hold for all $0 < t$.

The control scheme is designed according to the backstepping method, the following coordinate transformation is proposed:

$$\zeta_1 = \mathcal{Z}_1, \quad \zeta_j = \hat{x}_j - \nu_j, j = 2, \ldots, n \tag{22}$$

where ν_j is the output of the following command filter with respect to the virtual controller α_{j-1}

$$\varpi_j \dot{\nu}_j + \nu_j = \alpha_{j-1}, \nu_j(0) = \alpha_{j-1}(0) \tag{23}$$

with ϖ_j being the positive design constant. In addition, we define the filter error as $\varepsilon_j = \nu_j - \alpha_{j-1}$.

The controller will then be progressively designed using the backstepping technique.

Step 1: From (7) and (22), the derivative of ζ_1 is described as

$$\begin{aligned}\dot{\zeta}_1 =& \ \tau_1 \dot{z}_1 + \phi_1 \\ =& \ \tau_1\Big[\hat{x}_2 + \hat{\hbar}_1(\hat{\bar{x}}_2|\omega_1) + \ell_1(\bar{y} - \hat{y}) \ - \dot{y}_r\Big] + \phi_1 \\ =& \ \tau_1\Big[\zeta_2 + \alpha_1 + \varepsilon_2 - \tilde{\omega}_1^T S_1(\hat{\bar{x}}_2) + \omega_1^{*T} S_1(\hat{\bar{x}}_2) + \ell_1(\bar{y} - \hat{y}) \\ & - \dot{y}_r - \omega_1^{*T} S_1(\hat{x}_1) + \tilde{\omega}_1^T S_1(\hat{x}_1) + \omega_1^T S_1(\hat{x}_1)\Big] + \phi_1\end{aligned} \tag{24}$$

where

$$\tau_1 = \begin{cases} \frac{1}{\psi} - \frac{4bp}{d\psi^2}\left[\frac{\vartheta}{d} - 1\right]^{2p-1} z_1^2 > 0, 0 < \vartheta \le d \\ 1, \qquad\qquad\qquad\qquad \vartheta > d \end{cases} \tag{25}$$

$$\phi_1 = \begin{cases} \frac{4bp}{d\psi^2}\left[\frac{\vartheta}{d} - 1\right]^{2p-1} \dot{\eta}\eta z_1, 0 < \vartheta \le d, \\ 0, \qquad\qquad\qquad\qquad \vartheta > d. \end{cases} \tag{26}$$

The Lyapunov function V_1 is designed as

$$V_1 = V_0 + \ln\cosh(\zeta_1) + \frac{1}{2\lambda_1}\tilde{\omega}_1^T\tilde{\omega}_1 + \frac{1}{2\lambda_1^*}\tilde{\theta}_1^2 + \frac{1}{2}\varepsilon_2^2 \tag{27}$$

where λ_1 and λ_1^* are known positive constants. Let $\theta_1^* = \|\omega_1^*\|^2$, θ_1 is the estimation of θ_1^* with $\tilde{\theta}_1 = \theta_1^* - \theta_1$.

Then

$$\dot{V}_1 = \ \dot{V}_0 + \tanh(\zeta_1)\dot{\zeta}_1 + \frac{1}{\lambda_1}\tilde{\omega}_1^T\dot{\tilde{\omega}}_1 + \frac{1}{\lambda_1^*}\tilde{\theta}_1\dot{\tilde{\theta}}_1 + \varepsilon_2\dot{\varepsilon}_2. \tag{28}$$

With the help of Young's inequality, the following inequalities are established:

$$\begin{aligned}\tanh(\zeta_1)\tau_1\varepsilon_2 \le& \ \frac{1}{2}\tanh^2(\zeta_1)\tau_1^2 + \frac{1}{2}\varepsilon_2^2, \\ \tanh(\zeta_1)\phi_1 \le& \ r\tanh^2(\zeta_1)\phi_1^2 + \frac{1}{4r}, \\ \tanh(\zeta_1)\tau_1\ell_1(\bar{y} - \hat{y}) \le& \ 3r\tanh^2(\zeta_1)\tau_1^2 + \frac{1}{4r}\|\ell_1\|^2\varrho^2 \\ & + \frac{1}{4r}\|\ell_1\|^2\bar{\Lambda}^2\end{aligned} \tag{29}$$

and

$$\begin{aligned}&\tanh(\zeta_1)\tau_1(\omega_1^{*T} S_1(\hat{\bar{x}}_2) - \tilde{\omega}_1^T S_1(\hat{\bar{x}}_2) - \omega_1^{*T} S_1(\hat{x}_1)) \\ &\le \frac{2}{r} + \frac{1}{2}\tanh^2(\zeta_1)\tau_1^2 + \frac{1}{2}\tilde{\omega}_1^T\tilde{\omega}_1 + \frac{r}{2}\tanh^2(\zeta_1)\tau_1^2\theta_1^*\end{aligned} \tag{30}$$

where r is a known positive constant.

Combining (27), (28), and (29), we can get the following formula:

$$\begin{aligned}\dot{V}_1 \le& \ \dot{V}_0 + \tanh(\zeta_1)\Big[\tau_1(\zeta_2 + \alpha_1 + \frac{r}{\tau_1}\tanh(\zeta_1)\phi_1^2 \\ & + 3r\tanh(\zeta_1)\tau_1 + \tanh(\zeta_1)\tau_1) + \frac{r}{2}\tanh(\zeta_1)\tau_1\theta_1 \\ & - \dot{y}_r\Big] + \tilde{\omega}_1^T(\tanh(\zeta_1)\tau_1 S_1(\hat{x}_1) - \frac{1}{\lambda_1}\dot{\omega}_1) \\ & + \tilde{\theta}_1(\frac{r}{2}\tanh^2(\zeta_1)\tau_1^2 - \frac{1}{\lambda_1^*}\dot{\theta}_1) + \frac{1}{2}\varepsilon_2^2 + \frac{1}{4r}\|\ell_1\|^2\varrho^2 \\ & + \frac{1}{4r}\|\ell_1\|^2\bar{\Lambda}^2 + \frac{1}{2}\tilde{\omega}_1^T\tilde{\omega}_1 + \varepsilon_2\dot{\varepsilon}_2 + \frac{9}{4r}.\end{aligned} \tag{31}$$

The virtual controller α_1 as well as the parameter adaptive laws are structured as

$$\begin{aligned}\alpha_1 =& \ -\frac{\beta_1}{\tau_1}\tanh(\zeta_1) - \frac{1}{\tau_1}r\tanh(\zeta_1)\phi_1^2 - 3r\tanh(\zeta_1)\tau_1 \\ & - \tanh(\zeta_1)\tau_1 - \frac{r}{2}\tanh(\zeta_1)\tau_1\theta_1 + \dot{y}_r\end{aligned} \tag{32}$$

and

$$\begin{aligned}\dot{\theta}_1 &= \lambda_1^*\frac{r}{2}\tanh^2(\zeta_1)\tau_1^2 - s_1\theta_1 \\ \dot{\omega}_1 &= \lambda_1\tanh(\zeta_1)\tau_1 S_1(\hat{x}_1) - \kappa_1\omega_1\end{aligned} \tag{33}$$

with β_1, s_1 and κ_1 being positive constants to be designed.

Substituting (32) and (33) into (31) results in

$$\begin{aligned}\dot{V}_1 \le& \ -\rho_1\|\varrho\|^2 - \beta_1\tanh^2(\zeta_1) + \tau_1\tanh(\zeta_1)\zeta_2 \\ & + \frac{1}{c}\sum_{i=1}^{n}\|\tilde{\omega}_i\|^2 + \left[\frac{1}{c}\|\tilde{\omega}_g\|^2 + \varsigma\right]u^2 + \Theta_1 \\ & + \frac{\kappa_1}{\lambda_1}\tilde{\omega}_1^T\omega_1 + \frac{s_1}{\lambda_1^*}\tilde{\theta}_1\theta_1 + \frac{1}{2}\tilde{\omega}_1^T\tilde{\omega}_1 + \frac{1}{2}\varepsilon_2^2 + \varepsilon_2\dot{\varepsilon}_2\end{aligned} \tag{34}$$

where

$$\begin{aligned}\rho_1 &= \rho_0 - \frac{1}{4r}\|\ell_1\|^2 \\ \Theta_1 &= \Theta_0 + \frac{1}{4r}\|\ell_1\|^2\bar{\Lambda}^2 + \frac{9}{4r}.\end{aligned} \tag{35}$$

Introducing the Young's inequality, we have

$$\varepsilon_2\dot{\varepsilon}_2 \ \le \frac{1}{2}(\varepsilon_2)^2 + \frac{1}{2}(\dot{\alpha}_1)^2 - \frac{(\varepsilon_2)^2}{\varpi_2} \tag{36}$$

where

$$\dot{\alpha}_1 = \frac{\partial\alpha_1}{\partial\hat{x}_1}\dot{\hat{x}}_1 + \frac{\partial\alpha_1}{\partial\theta_1}\dot{\theta}_1 + \frac{\partial\alpha_1}{\partial\omega_1}\dot{\omega}_1 + \sum_{i=0}^{1}\frac{\partial\alpha_1}{\partial y_r^i}y_r^{(i+1)}.$$

Through the above analysis, we select $(1/\varpi_2) \ge 1 + b_2$ with $b_2 > 0$ being a design parameter. Then, it can be obtained that

$$\begin{aligned}\dot{V}_1 \le& \ -\rho_1\|\varrho\|^2 - \beta_1\tanh^2(\zeta_1) + \tau_1\tanh(\zeta_1)\zeta_2 - b_2(\varepsilon_2)^2 \\ & + \left[\frac{1}{c}\|\tilde{\omega}_g\|^2 + \varsigma\right]u^2 + \Theta_1 + \frac{1}{c}\sum_{i=1}^{n}\|\tilde{\omega}_i\|^2 + \frac{1}{2}\dot{\alpha}_1^2 \\ & + \frac{\kappa_1}{\lambda_1}\tilde{\omega}_1^T\omega_1 + \frac{s_1}{\lambda_1^*}\tilde{\theta}_1\theta_1 + \frac{1}{2}\tilde{\omega}_1^T\tilde{\omega}_1.\end{aligned} \tag{37}$$

Step j (2 ≤ j ≤ n − 1): According to (22), we can obtain the derivative of ζ_j as

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$$\begin{aligned}\dot{\zeta}_j =& \dot{\hat{x}} - \dot{\vartheta}_j \\ =& \zeta_{j+1} + \alpha_j + \varepsilon_{j+1} - \tilde{\omega}_j^T S_j(\hat{\bar{x}}_{j+1}) + \omega_j^{*T} S_j(\hat{\bar{x}}_{j+1}) \\ &+ \ell_{j+1}(\bar{y} - \hat{y}) \ - \dot{\vartheta}_j - \omega_j^{*T} S_j(\hat{\bar{x}}_j) + \omega_j^T S_j(\hat{\bar{x}}_j) \\ &+ \tilde{\omega}_j^T S_j(\hat{\bar{x}}_j). \end{aligned} \tag{38}$$

Construct the Lyapunov candidate function as

$$V_j = V_{j-1} + \ln\cosh(\zeta_j) + \tfrac{1}{2\lambda_j}\tilde{\omega}_j^T\tilde{\omega}_j + \tfrac{1}{2\lambda_j^*}\tilde{\theta}_j^2 + \tfrac{1}{2}\varepsilon_{j+1}^2$$

where λ_j and λ_j^* are known positive constants. Defining $\theta_j^* = \|\omega_j^*\|^2$, θ_j is the estimation of θ_j^* with $\tilde{\theta}_j = \theta_j^* - \theta_j$.

It follows that

$$\begin{aligned}\dot{V}_j \le& -\rho_{j-1}\|\varrho\|^2 - \sum_{i=1}^{j-1}\beta_i\tanh^2(\zeta_i) + \tau_1\tanh(\zeta_1)\zeta_2 \\ &+ \sum_{i=1}^{j-1}\frac{s_i}{\lambda_i^*}\tilde{\theta}_i\theta_i + \sum_{i=1}^{j-1}\frac{1}{2}\tilde{\omega}_i^T\tilde{\omega}_i + \Theta_{j-1} + \sum_{i=1}^{j-1}\frac{1}{2}\dot{\alpha}_i^2 \\ &+ \sum_{i=2}^{j-1}\tanh(\zeta_i)\zeta_{i+1} - \sum_{i=2}^{j} b_i(\varepsilon_i)^2 + \sum_{i=1}^{j-1}\frac{\kappa_i}{\lambda_i}\tilde{\omega}_i^T\omega_i \\ &+ \frac{1}{c}\sum_{i=1}^{n}\|\tilde{\omega}_i\|^2 + \left[\frac{1}{c}\|\tilde{\omega}_g\|^2 + \varsigma\right]u^2 + \tanh(\zeta_j)\big(\alpha_j \\ &+ \zeta_{j+1} + \varepsilon_{j+1} - \tilde{\omega}_j^T S_j(\hat{\bar{x}}_{j+1}) + \omega_j^{*T} S_j(\hat{\bar{x}}_{j+1}) \\ &+ \ell_{j+1}(\bar{y} - \hat{y}) \ - \dot{\vartheta}_j - \omega_j^{*T} S_j(\hat{\bar{x}}_j) + \omega_j^T S_j(\hat{\bar{x}}_j)\big) \\ &+ \tilde{\omega}_j^T S_j(\hat{\bar{x}}_j) + \varepsilon_{j+1}\dot{\varepsilon}_{j+1} + \frac{1}{\lambda_j}\tilde{\omega}_j^T\dot{\tilde{\omega}}_j + \frac{1}{\lambda_j^*}\tilde{\theta}_j\dot{\tilde{\theta}}_j \end{aligned} \tag{39}$$

where

$$\Theta_{j-1} = \Theta_1 + \sum_{i=2}^{j-1}\frac{1}{4r}\|\ell_i\|^2\bar{\Lambda}^2 + \frac{2j-2}{r}. \tag{40}$$

In addition, similar to Step 1, the virtual controller α_j is designed as

$$\begin{aligned}\alpha_j =& -\beta_j\tanh(\zeta_j) - 3r\tanh(\zeta_j) - \tanh(\zeta_j) \\ &- \frac{r}{2}\tanh(\zeta_j)\theta_j + \dot{\vartheta}_j \end{aligned} \tag{41}$$

and the parameter adaptive laws are constructed as

$$\begin{aligned}\dot{\theta}_j &= \lambda_j^*\frac{r}{2}\tanh^2(\zeta_j) - s_j\theta_j \\ \dot{\omega}_j &= \lambda_j\tanh(\zeta_j)S_j(\hat{\bar{x}}_j) - \kappa_j\omega_j \end{aligned} \tag{42}$$

where $\beta_j > 0$, $s_j > 0$ and $\kappa_j > 0$ are know constants.

In addition, we can obtain

$$\varepsilon_{j+1}\dot{\varepsilon}_{j+1} \ \le -\frac{(\varepsilon_{j+1})^2}{\varpi_{j+1}} + \frac{1}{2}(\varepsilon_{j+1})^2 + \frac{1}{2}(\dot{\alpha}_j)^2 \tag{43}$$

where

$$\dot{\alpha}_j = \sum_{i=1}^{j}\frac{\partial\alpha_j}{\partial\hat{x}_i}\dot{\hat{x}}_i + \frac{\partial\alpha_j}{\partial\omega_i}\dot{\omega}_i + \frac{\partial\alpha_j}{\partial\theta_i}\dot{\theta}_i + \sum_{i=0}^{j}\frac{\partial\alpha_j}{\partial y_r^i}y_r^{(i+1)}.$$

Here, we select $(1/\varpi_{j+1}) \ge 1 + b_{j+1}$ with $b_{j+1} > 0$ being a design parameter. Thus, the derivative of V_j becomes

$$\begin{aligned}\dot{V}_j \le& -\rho_j\|\varrho\|^2 - \sum_{i=1}^{j}\beta_i\tanh^2(\zeta_i) - \sum_{i=2}^{j+1}b_i(\varepsilon_i)^2 \\ &+ \sum_{i=1}^{j}\frac{s_i}{\lambda_i^*}\tilde{\theta}_i\theta_i + \sum_{i=1}^{j}\frac{\kappa_i}{\lambda_i}\tilde{\omega}_i^T\omega_i + \frac{1}{c}\sum_{i=1}^{n}\|\tilde{\omega}_i\|^2 \\ &+ \tau_1\tanh(\zeta_1)\zeta_2 + \sum_{i=2}^{j}\tanh(\zeta_i)\zeta_{i+1} + \sum_{i=1}^{j}\frac{1}{2}\dot{\alpha}_i^2 \\ &+ \left[\frac{1}{c}\|\tilde{\omega}_g\|^2 + \varsigma\right]u^2 + \sum_{i=1}^{j}\frac{1}{2}\tilde{\omega}_i^T\tilde{\omega}_i + \Theta_j \end{aligned} \tag{44}$$

where

$$\begin{aligned}\rho_j &= \rho_{j-1} - \frac{1}{4r}\|\ell_j\|^2 \\ \Theta_j &= \Theta_{j-1} + \frac{1}{4r}\|\ell_j\|^2\bar{\Lambda}^2 + \frac{2}{r}. \end{aligned} \tag{45}$$

Step n: In order to ensure that the controller u being designed in the last step is bounded, hence, we introduce the concept of the bounded control design idea [37].

From (7) and (22), the derivative of ζ_n can be expressed as

$$\dot{\zeta}_n = \hat{\mathfrak{g}}_n(\hat{\bar{x}}_n|\omega_n)u + \hat{h}_n(\hat{\bar{x}}_n|\omega_n) + \ell_n(\bar{y} - \hat{y}) \ - \dot{\vartheta}_n. \tag{46}$$

We construct the following Lyapunov function V_n:

$$V_n = V_{n-1} + \ln\cosh(\zeta_n) + \frac{1}{2\lambda_n}\tilde{\omega}_n^T\tilde{\omega}_n + \frac{1}{2\lambda_g^*}\tilde{\omega}_g^T\tilde{\omega}_g \tag{47}$$

where λ_n and λ_g^* are known positive constants.

Then

$$\begin{aligned}\dot{V}_n \le& \ \dot{V}_{n-1} + \tanh(\zeta_n)\big(\omega_g^T S_g(\hat{\bar{x}}_n)u + \omega_n^T S_n(\hat{\bar{x}}_n) \\ &+ \ell_n(\bar{y} - \hat{y}) \ - \dot{\vartheta}_n - \tilde{\omega}_g^T S_g(\hat{\bar{x}}_n)u - \tilde{\omega}_n^T S_n(\hat{\bar{x}}_n)\big) \\ &+ \tilde{\omega}_n^T(\tanh(\zeta_n)S_n(\hat{\bar{x}}_n) - \frac{1}{\lambda_n}\dot{\omega}_n) \\ &+ \tilde{\omega}_g^T(\tanh(\zeta_n)S_g(\hat{\bar{x}}_n)u - \frac{1}{\lambda_n^*}\dot{\omega}_g) \end{aligned} \tag{48}$$

where $\tilde{\omega}_n = \omega_n^* - \omega_n$ and $\tilde{\omega}_g = \omega_g^* - \omega_g$.

By introducing the Young's inequality, we get

$$\begin{aligned} -\tanh(\zeta_n) \quad & (\tilde{\omega}_g^T S_g(x_n)u + \tilde{\omega}_n^T S_n(x_n)) \\ &\le \tanh^2(\zeta_n) + \tfrac{1}{2}\|\tilde{\omega}_n\|^2 + \tfrac{1}{2}\|\tilde{\omega}_g\|^2u^2 \\ \tanh(\zeta_n) \quad & \ell_n(\bar{y} - \hat{y}) \\ &\le 3r\tanh^2(\zeta_n) + \tfrac{1}{4r}\|\ell_n\|^2\varrho^2 \\ &\quad + \tfrac{1}{4r}\|\ell_n\|^2\bar{\Lambda}^2. \end{aligned} \tag{49}$$

Combining (48), (49), we get

$$\begin{aligned}\dot{V}_n \le& -\rho_n\|\varrho\|^2 - \sum_{i=1}^{n-1}\beta_i\tanh^2(\zeta_i) + \tau_1\tanh(\zeta_1)\zeta_2 \\ &+ \left[\frac{1}{c}\|\tilde{\omega}_g\|^2 + \varsigma\right]u^2 + \frac{1}{c}\sum_{i=1}^{n}\|\tilde{\omega}_i\|^2 \\ &+ \sum_{i=1}^{n-1}\frac{s_i}{\lambda_i^*}\tilde{\theta}_i\theta_i + \sum_{i=1}^{n-1}\frac{1}{2}\tilde{\omega}_i^T\tilde{\omega}_i + \Theta_{n-1} + \sum_{i=1}^{n-1}\frac{1}{2}\dot{\alpha}_i^2 \end{aligned}$$

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$$+\sum_{i=2}^{n-1}\tanh(\zeta_i)\zeta_{i+1}-\sum_{i=2}^{n}b_i(\varepsilon_i)^2+\sum_{i=1}^{n-1}\frac{\kappa_i}{\lambda_i}\tilde{\omega}_i^T\omega_i$$
$$+\tanh(\zeta_n)\big(\omega_g^T S_g(\hat{\bar{x}}_n)u+\omega_n^T S_n(\hat{\bar{x}}_n)-\dot{\vartheta}_n$$
$$+\tanh(\zeta_n)+3r\tanh(\zeta_n)\big)+\frac{1}{2}\|\tilde{\omega}_g\|^2u^2$$
$$+\frac{1}{2}\|\tilde{\omega}_n\|^2+\tilde{\omega}_n^T(\tanh(\zeta_n)S_n(\hat{\bar{x}}_n)-\frac{1}{\lambda_n}\dot{\omega}_n)$$
$$+\tilde{\omega}_g^T(\tanh(\zeta_n)S_g(\hat{\bar{x}}_n)u-\frac{1}{\lambda_n^*}\dot{\omega}_g) \tag{50}$$

where

$$\rho_n=\rho_{n-1}-\frac{1}{4r}\|\ell_n\|^2$$
$$\Theta_n=\Theta_{n-1}+\frac{1}{4r}\|\ell_n\|^2\bar{\Lambda}^2. \tag{51}$$

Thus, (50) can be rewritten as

$$\dot{V}_n\le\ -\rho_n\|\varrho\|^2-\sum_{i=1}^{n-1}\beta_i\tanh^2(\zeta_i)-\sum_{i=2}^{n}b_i(\varepsilon_i)^2$$
$$+\sum_{i=1}^{n-1}\frac{\kappa_i}{\lambda_i}\tilde{\omega}_i^T\omega_i+\sum_{i=1}^{n-1}\frac{s_i}{\lambda_i^*}\tilde{\theta}_i\theta_i+\sum_{i=1}^{n-1}\frac{1}{2}\tilde{\omega}_i^T\tilde{\omega}_i$$
$$+\Theta_n+\sum_{i=1}^{n-1}\frac{1}{2}\dot{\alpha}_i^2+\frac{\tau_1^2}{2}\zeta_1^2+\sum_{i=2}^{n-1}\frac{1}{2}\zeta_i^2+\sum_{i=1}^{n-1}\frac{1}{2}\zeta_{i+1}^2$$
$$+\frac{1}{c}\sum_{i=1}^{n}\|\tilde{\omega}_i\|^2+\left[(\frac{1}{c}+\frac{1}{2})\|\tilde{\omega}_g\|^2+\varsigma\right]u^2$$
$$+\tanh(\zeta_n)\big(\omega_g^T S_g(\hat{\bar{x}}_n)u+\omega_n^T S_n(\hat{\bar{x}}_n)-\dot{\vartheta}_n$$
$$+\tanh(\zeta_n)+3r\tanh(\zeta_n)\big)+\frac{1}{2}\|\tilde{\omega}_n\|^2$$
$$+\tilde{\omega}_n^T(\tanh(\zeta_n)S_n(\hat{\bar{x}}_n)-\frac{1}{\lambda_n}\dot{\omega}_n)$$
$$+\tilde{\omega}_g^T(\tanh(\zeta_n)S_g(\hat{\bar{x}}_n)u-\frac{1}{\lambda_n^*}\dot{\omega}_g). \tag{52}$$

The controller u is designed as

$$u=\frac{-(\beta_n+1+3r)\tanh(\zeta_n)-\omega_n^T S_n(\hat{\bar{x}}_n)+\dot{\vartheta}_n}{\omega_g^T S_g(\hat{\bar{x}}_n)} \tag{53}$$

and the parameter adaptive laws are constructed as

$$\dot{\omega}_g=\lambda_g^*\tanh(\zeta_n)S_g(\hat{\bar{x}}_n)u-s_g\omega_g$$
$$\dot{\omega}_n=\lambda_n\tanh(\zeta_n)S_n(\hat{\bar{x}}_n)-\kappa_n\omega_n \tag{54}$$

where $\beta_n>0$, $s_g>0$ and $\kappa_n>0$ are know constants.

According to (53), it is clear that if ω_g, ω_n are bounded and $\omega_g\neq 0$, it follows that the controller u is bounded. In order to be able to guarantee the parameters ω_g and ω_n always keep in the predefined bounded sets, we resort to projection algorithm [38] to aid in the construction of adaptive parameter laws

$$\dot{\omega}_g=\lambda_g^* Proj\Big[\omega_g,\tanh(\zeta_n)S_g(\hat{\bar{x}}_n)u-s_g\omega_g\Big]$$
$$\dot{\omega}_n=\lambda_n Proj\Big[\omega_n,\tanh(\zeta_n)S_n(\hat{\bar{x}}_n)-\kappa_n\omega_n\Big]. \tag{55}$$

In addition, if $\omega_g(0)>0$ and $\omega_n(0)>0$ are satisfied, the parameter adaptive laws are able to guarantee $\omega_g(t)$ and $\omega_n(t)$ to remain positive and bounded. Therefore, we suppose $\|\tilde{\omega}_g\|\le\bar{\omega}_g$, $|u|<\bar{u}$.

Inserting (53) and (55) into (52) yields the following inequality

$$\dot{V}_n\le\ -\rho_n\|\varrho\|^2-\sum_{i=1}^{n}\beta_i\tanh^2(\zeta_i)-\sum_{i=2}^{n}b_i(\varepsilon_i)^2$$
$$+\frac{1}{c}\sum_{i=1}^{n}\|\tilde{\omega}_i\|^2+\left[(\frac{1}{c}+\frac{1}{2})\|\tilde{\omega}_g\|^2+\varsigma\right]u^2$$
$$+\sum_{i=1}^{n-1}\frac{1}{2}\tilde{\omega}_i^T\tilde{\omega}_i+\sum_{i=1}^{n-1}\frac{s_i}{\lambda_i^*}\tilde{\theta}_i\theta_i+\sum_{i=1}^{n}\frac{\kappa_i}{\lambda_i}\tilde{\omega}_i^T\omega_i$$
$$+\sum_{i=1}^{n-1}\frac{1}{2}\dot{\alpha}_i^2+\frac{\tau_1^2}{2}\zeta_1^2+\sum_{i=2}^{n-1}\frac{1}{2}\zeta_i^2+\sum_{i=1}^{n-1}\frac{1}{2}\zeta_{i+1}^2$$
$$+\frac{1}{2}\|\tilde{\omega}_n\|^2+\frac{s_g}{\lambda_g^*}\tilde{\omega}_g^T\omega_g+\Theta_n. \tag{56}$$

C. System Stability

Theorem 1: Considering the uncertain nonlinear system (1) that satisfies Assumption 1, if the adaptive controller (53) is applied, and the parameter adaptive laws are presented by (33), (42), and (55), thus it can be concluded that all the closed-loop system signals as well as the observer estimation error are ensured to be bounded by adjusting suitable parameters. Moreover, there is a positive constant $t^{\circledcirc}$ for all $k=0,1\ldots$ the interexecution intervals $t_{y,k+1}-t_{y,k}\ge t^{\circledcirc}$ are satisfied.

Proof: From (56), we can obtain the following inequalities:

$$\tilde{\omega}_i^T\omega_i\ \le-\frac{1}{2}\tilde{\omega}_i^T\tilde{\omega}_i+\frac{1}{2}\omega_i^{*T}\omega_i^*$$
$$\tilde{\omega}_g^T\omega_g\ \le-\frac{1}{2}\tilde{\omega}_g^T\tilde{\omega}_g+\frac{1}{2}\omega_g^{*T}\omega_g^*$$
$$\tilde{\theta}_i\theta_i\ \le-\frac{1}{2}\tilde{\theta}_i^2+\frac{1}{2}\theta_i^{*2}. \tag{57}$$

After the previous analyses, it is clear that the set $\bar{\Omega}:=\{V_n|V_n\le\Xi_2(||\mathcal{J}||)\}$ ($\Xi_2(||\mathcal{J}||)$ will be introduced in the following) is compact in R^{2n+2}. Further, $|\dot{\alpha}_i|\le\Delta$ on $\Xi_2(||\mathcal{J}||)$ with $i=1,\ldots,n-1$. To make the stability analysis more convenient, we define δ, π, φ_i, b_i^* and ϱ_i ($i=1,\ldots,n$) to be positive constants, and $\bar{\rho}_n=\rho_n+\delta$, $\bar{\pi}=(s_g/2\lambda_g^*)-1/c+\pi$, $\bar{\varphi}_i=(\kappa_i/2\lambda_i)-1/2-1/c+\varphi_i$, $\tilde{b}_i=b_i-1/c+b_i^*$, $\bar{\varrho}_i=(s_i/2\lambda_i^*)-1/c+\varrho_i$. Then, we have

$$\dot{V}_n\le\ -\bar{\rho}_n\|\varrho\|^2-\sum_{i=1}^{n}\beta_i\tanh^2(\zeta_i)-\sum_{i=1}^{n}\bar{\varphi}_i\tilde{\omega}_i^T\tilde{\omega}_i$$
$$-\sum_{i=1}^{n-1}\bar{\varrho}_i\tilde{\theta}_i\tilde{\theta}_i-\bar{\pi}\tilde{\omega}_g^T\tilde{\omega}_g-\sum_{i=2}^{n}\tilde{b}_i(\varepsilon_i)^2$$
$$+\delta\|\varrho\|^2+\frac{\tau_1^2}{2}\zeta_1^2+\sum_{i=2}^{n}\zeta_i^2+\sum_{i=1}^{n}\varphi_i\tilde{\omega}_i^T\tilde{\omega}_i$$
$$+\sum_{i=1}^{n-1}\varrho_i\tilde{\theta}_i\tilde{\theta}_i+\pi\tilde{\omega}_g^T\tilde{\omega}_g+\sum_{i=2}^{n}b_i^*(\varepsilon_i)^2$$
$$+\bar{\Theta}_n \tag{58}$$

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where $\bar{\Theta}_n = \Theta_n + (1/2)\Delta^2 + \sum_{i=1}^{n}[\kappa_i/(2\lambda_i)]\omega_i^{*T}\omega_i^* + \sum_{i=1}^{n-1}[s_i/(2\lambda_i^*)]\theta_i^{*2} + [s_g/(2\lambda_g^*)]\omega_g^{*T}\omega_g^* + [([1/c] + [1/2])\|\tilde{\omega}_g\|^2 + \varsigma]\bar{u}^2$.

The following two vectors are defined:

$$\mathcal{J}^T = \left[\varrho, \bar{\zeta}_n, \bar{\varepsilon}, \bar{\tilde{\omega}}, \bar{\tilde{\theta}}, \tilde{\omega}_g\right]$$
$$\mathcal{K}^T = \left[\varrho, \tanh(\bar{\zeta}_n), \bar{\varepsilon}, \bar{\tilde{\omega}}, \bar{\tilde{\theta}}, \tilde{\omega}_g\right] \tag{59}$$

where $\bar{\zeta}_n = [\zeta_1, \ldots, \zeta_n]^T$, $\tanh(\bar{\zeta}_n) = [\tanh(\zeta_1), \ldots, \tanh(\zeta_n)]^T$, $\bar{\varepsilon} = [\varepsilon_2, \ldots, \varepsilon_n]^T$, $\bar{\tilde{\omega}} = [\tilde{\omega}_1, \ldots, \tilde{\omega}_n]^T$, and $\bar{\tilde{\theta}} = [\tilde{\theta}_1, \ldots, \tilde{\theta}_{n-1}]^T$.

Immediately after that, we can organize (58) into the following form:

$$\dot{V}_n \le \beta_1^*||\mathcal{J}||^2 - \beta_2^*||\mathcal{K}||^2 + \bar{\Theta}_n \tag{60}$$

where $\beta_1^* = \min\{\bar{\rho}_n, \beta_i, \bar{\varphi}_i, \tilde{b}_i, \bar{\varrho}_i, \bar{\pi}\}$ and $\beta_2^* = \min\{\delta, (\tau_1^2/2), 1, \varphi_i, b_i^*, \varrho_i, \pi\}$.

Further, if

$$\beta_1^*||\mathcal{J}||^2 - \beta_2^*||\mathcal{K}||^2 \le 0 \tag{61}$$

is satisfied, (60) can be rewritten as

$$\dot{V}_n \le -\beta_3^*||\mathcal{K}||^2 + \bar{\Theta}_n \tag{62}$$

where β_3^* is the positive constant.

Based on $||\mathcal{K}||^2 \ge \tanh^2(||\mathcal{J}||)$ [37], [38] and Lemma 1, the sufficient condition for (61) is guaranteed by the following:

$$(\|\mathcal{J}\| + 1)^2 \le \frac{\beta_2^*}{\beta_1^*}. \tag{63}$$

In addition, by (47), it is known that the following property is satisfied:

$$\Xi_1(\|\mathcal{J}\|) \le V_n \le \Xi_2(\|\mathcal{J}\|) \tag{64}$$

where $\Xi_1(\|\mathcal{J}\|) = \frac{1}{2}\ln(\cosh(\mathcal{J}))$ and $\Xi_2(\|\mathcal{J}\|) = \|\mathcal{J}\|^2$.

This also means that

$$\|\mathcal{J}\| \le \cosh^{-1}(\exp(2V_n)). \tag{65}$$

In turn, when the following condition holds:

$$(\cosh^{-1}(\exp(2V_n)) + 1)^2 \le \frac{\beta_2^*}{\beta_1^*} \tag{66}$$

we are able to ensure that (63) is established. It is worth noting that (66) can be satisfied by the sufficient condition defined in (69).

With the help of $||\mathcal{K}||^2 \ge \tanh^2(||\mathcal{J}||)$, the inequality (62) is simplified as

$$\dot{V}_n \le -\Xi_3(||\mathcal{J}||) + \bar{\Theta}_n \tag{67}$$

where $\Xi_3(||\mathcal{J}||) = \beta_3^*\tanh^2(||\mathcal{J}||)$. This implies that $||\mathcal{J}||$ is uniformly ultimately bound [37], and

$$||\mathcal{J}|| \le \bar{\mathcal{J}} \ \forall t^* \ge T^*(\bar{\mathcal{J}}, ||\mathcal{J}(0)||) \tag{68}$$

where $\mathcal{J}$ is the radius of the sphere, including ϱ, $\bar{\zeta}_j$, $\bar{\varepsilon}_{i+1}$, $\bar{\tilde{\omega}}_i$, $\bar{\tilde{\theta}}_i$ and $\tilde{\omega}_g$. The ultimate time to arrive at the sphere is $T^*(\bar{\mathcal{J}}, ||\mathcal{J}(0)||) > 0$.

With the help of the monotonically increasing nature of the $\cosh^{-1}(\diamond)$ function, the sufficient condition for (61) can be derived from (64) and (68) by selecting suitable β_1^* and β_2^*

$$(\cosh^{-1}(\exp(2\mathcal{I}^2)) + 1)^2 \le \frac{\beta_2^*}{\beta_1^*} \tag{69}$$

where $\mathcal{I} = \max(\bar{\mathcal{J}}, ||\mathcal{J}(0)||)$. Furthermore, $\bar{\mathcal{J}}$ and $T^*(\bar{\mathcal{J}}, ||\mathcal{J}(0)||) > 0$ are constructed as

$$\bar{\mathcal{J}} > (\Xi_1^{-1} \circ \Xi_2)(\Xi_3^{-1}(\bar{\Theta}_n))$$
$$T^* = \begin{cases} 0, & \text{if } \|\mathcal{K}(0)\| \le (\Xi_2^{-1} \circ \Xi_1)(\bar{\mathcal{J}}), \\ \frac{\Xi_2(\|\mathcal{K}(0)\|) - \Xi_1((\Xi_2^{-1}\circ\Xi_1)(\bar{\mathcal{J}}))}{\Xi_3(\Xi_2^{-1}\circ\Xi_1)(\bar{\mathcal{J}}) - \bar{\Theta}_n}, & \text{otherwise.} \end{cases} \tag{70}$$

According to (68), it is known that all closed-loop signals are proven to be bounded by $2\bar{\mathcal{J}}$ as well as the observer estimation error are bounded. Since ζ_1 is bounded, $-\eta(t) \le z_1(t) \le \eta(t)$ holds, and then by definition of $\eta(t)$, we finally get $|z_1| < m$ for $t > T$.

To exclude Zeno behavior, we will prove that the lower bound of the execution interval is a positive constant. For $t \in [t_{y,k}, t_{y,k+1})$, let $e^*(t) = y(t) - \bar{y}(t)$. During this time interval, the magnitude of $|e^*|$ initially begins at zero and may subsequently return to zero multiple times. Define $t_{y,l}$ as the last instant when $|e^*|$ hits zero, thus $t_{y,k} < t_{y,l} < t_{y,k+1}$ is satisfied. Meanwhile, we can get $|e^*| > 0$ in the interval $(t_{y,l}, t_{y,k+1})$. Therefore, we have

$$\frac{d}{dt}|e^*| = \frac{d}{dt}\sqrt{e^{*2}} = \text{sign}(y(t) - \bar{y}(t))\dot{y} \le |\dot{y}|, t \in [t_{y,l}, t_{y,k+1}). \tag{71}$$

Based on system (1), we can obtain $\dot{y} = \mathsf{g}_1 x_2 + f_1$. Since x_2, g_1 and f_1 are bounded, there is certainly a positive constant K so that $|\dot{y}| \le |\mathsf{g}_1 x_2 + f_1| \le K$. From $e^*(t_{y,l}) = 0$, we can obtain the lower bound of the execution interval as $t_{y,k+1} - t_{y,k} = t_{y,k+1} - t_{y,l} + t_{y,l} - t_{y,k} \ge \Lambda/K + t_{y,l} - t_{y,k}$. Thereby, the Zeno behavior is avoided. Hence, this accomplishes the proof of Theorem 1. ■

IV. Simulation Studies

In this study, we adopt the robotic manipulator system (see Fig. 1) described in [39] to illustrate the feasibility of our proposed control scheme. The dynamics of this system adhere to the Lagrangian equation form

$$Q\ddot{q}(t) + H\dot{q}(t) + MgL\sin(q(t)) = u \tag{72}$$

where the symbol Q represents the rotational inertia of the servo motor, while q and $\dot{q}$ denote the angle and angular velocity of the rigid link, respectively. H stands for damping coefficient. L is defined as the distance from the axis of the joint to the centre of mass. M is the mass of the link and g denotes the acceleration due to gravity. In the simulation of this article, the parameters have been chosen as $Q = 1$ $(\text{kg} \cdot \text{m}^2)$, $H = 2$ $(N \cdot s/m)$ and $MgL = 10$ $(N \cdot m)$. Further, we define $x_1 = q(t)$ and the desired trajectory $y_r(t) = 0.5\ \cos(t)$.

In the simulation, the initial values are set as $q(0) = 1$ and $\dot{q}(0) = 0$. The desired tracking accuracy $m = 3.5$ and the

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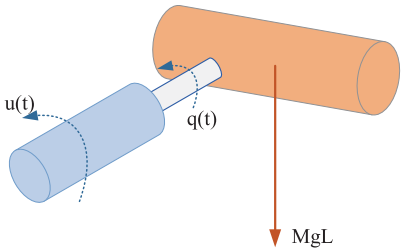


Fig. 1. Single-link robot arm.

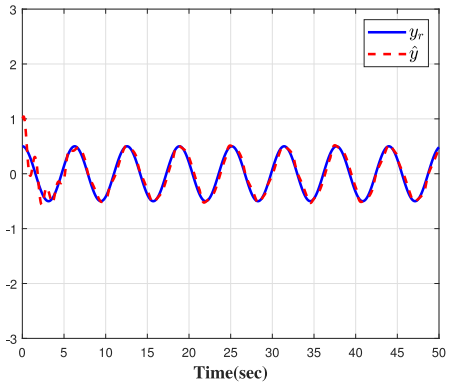


Fig. 2. Trajectories of $\hat{y}$ and y_r.

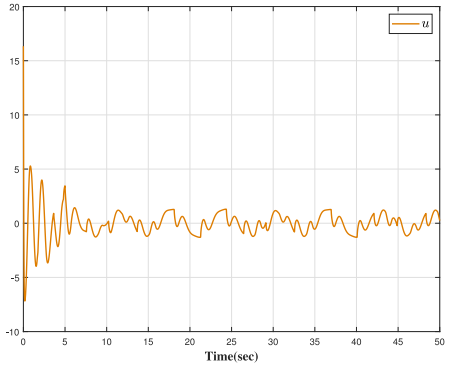


Fig. 3. Trajectoriy of u.

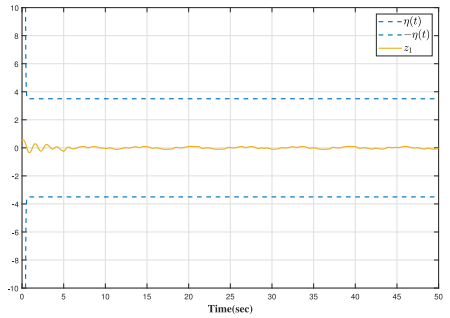


Fig. 4. Trajectory of tracking error z_1.

predefined time $T = 1s$.Other relevant parameters are selected as $\beta_1 = \beta_2 = 30$, $\ell_1 = \ell_2 = 2$, $r = 0.1$ and $\Lambda = 0.1$.

The simulation results regarding the single-link robot arm are presented in Figs. 2–5, respectively. Among them, Fig. 2 gives the curve of y_r and $\hat{y}$. The curve of the controller u is represented by Fig. 3. Fig. 4 depicts the tracking error.

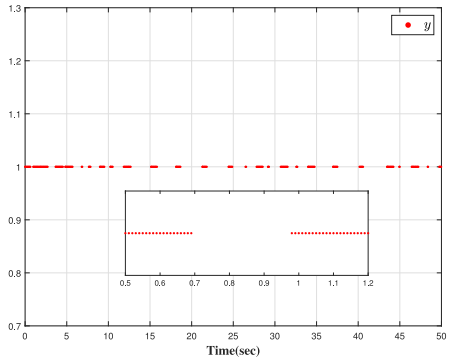


Fig. 5. Triggering times of the sensor output.

The result shows that $|z_1(t)| < m$ always holds at $t \geq T$. Additionally, the triggering times of the sensor output are depicted in Fig. 5, indicating that the interexecution interval remains strictly positive. Moreover, during this simulation, the total number of triggers is 1451, and further calculation shows that the saving rate of communication resources is 70.98%. This proves the validity of event triggering and saves communication resources. These simulation results unequivocally demonstrate that the constructed control scheme guarantees prescribed-time tracking control.

V. Conclusion

In this study, the adaptive prescribed-time tracking control problem has been investigated for a class of strict-feedback nonlinear systems with unknown control gains. The addition of the event-triggering mechanism in the communication channel from the sensor to the controller has reduced communication overhead and enhanced resource utilization. By utilizing triggered output signal and FLSs, a novel fuzzy state observer has been constructed to generate continuous states that can substitute for the original states. At the same time, the challenge of virtual signals being nondifferentiable has been solved by introducing the command filtering technology. Moreover, by establishing a time-varying constraint function and the logarithmic Lyapunov functions, and combining them with the backstepping design methodology as well as the bounded control technique, a prescribed-time fuzzy controller is designed, which achieves tracking within the prescribed time and ensures the boundedness of all signals in the closed-loop system. The simulation results have validated the rationality and effectiveness of the control scheme. The future research direction involves applying the proposed control scheme to the nonlinear heterogeneous MASs.

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